Operating Systems LAB (LAB-217)
Credit Hrs: **LAB 1 (Pre-Req : Data Structures)**
Lab 3 : **More Linux Commands**
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Semester: **Fall 2025**
Section : **C**
09-October, 2025
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Objectives:

The objective of this lab session is to introduce students to the Linux command-line interface (CLI) and help them become comfortable with its basic usage. Through hands-on practice, students will learn to perform essential file and directory management tasks, record and review their own terminal sessions, and appreciate how the CLI can often be faster, more efficient, and more powerful than graphical interfaces. Although we're primarily using Ubuntu, most of the commands explored in this session are fundamental to all Unix-like systems, including macOS, making the skills learned here widely applicable.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- **PLO5: Modern Tool Usage**
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mapped PLOs
CLO1	**Practice** Linux commands and utilities to gain proficiency with the command-line interface and use it effectively for system management tasks, basic scripting, software installation, and other essential operations in a Linux environment.	A5	PLO5

Practice Exercises

Before we begin today's graded lab exercise:

- Open your **terminal** as we start the session.
- Follow along and replicate everything I do and try to internalize the commands and concepts.
- This will prepare you for today's **graded exercise**.

We practiced following commands in previous week ;

`echo, whoami, hostname, pwd, cd, ls, clear, script, touch, tee, mkdir`

and explored their powers by using a lot of cool switches along with them and piping together multiple commands to perform multiple things as a single unit to do a lot of useful activities. These commands will still come handy today and might help you a lot in performing today's graded task.

These are the commands we are planning to learn and use in today's lab session ;

`which, whatis, man, help, wc, cp, mv, head, tail, more, less, grep`

and of course we'll see their variants and use cases by learning about different possible switches and piping these commands together to perform useful activities.

We will continue playing around with files and learn about copying, moving and renaming files. We'll play around with real world data and perform very basic yet very useful data analysis by piping together different commands as a single unit and honing their strengths with use of **switches**.

Your lab task for today is primarily based upon these things so make sure to practice with me so you can easily attempt your lab task.

Graded Exercise

- Open your terminal and navigate to the directory you want to work in (preferably Desktop) and remove unnecessary files from your working directory so it won't create troubles for you later on.

- Display a short introduction message containing your name and roll number on the terminal.

- **Create** a directory named after your own name.

- Download the **OS-Lab-Students.csv** file shared with you on the GCR. Make a copy of it in your **User's home directory** and name it **OS-Lab-Students-C.csv** and list out the contents of that directory for confirmation.

- **Move** that copied file in that directory which is named after your own name.

- Navigate into that directory named after you and display its content for verification that the file is moved successfully and is now available in this directory.

- Now search within that file (using **grep** command) those lines which contain **your name.** You also need to fetch and store the **line number** at which your name appeared in this file (Grep has that power) , and write the results into a new csv file named after **your name**. You'll be doing it by piping together 2 commands (You can use **tee** command for writing into a file). Now, display the content of this file.

- Similarly, search for those lines containing **Section-B** and write the results into a file named **Section-B**.csv. Pipe it with a 3rd command to fetch and display the number of lines of this newly created file (use **wc** command and its relevant switch). Now, display the first 5 lines of **Section-B.csv** using **head** command and its relevant switch.

- Similarly, search for those lines containing **Section-C** and write the results into a file named **Section-C**.csv. Pipe it with a 3rd command to fetch and display the number of lines of this newly created file (use **wc** command and its relevant switch). Now, display the first 5 lines of **Section-C.csv** using **head** command and its relevant switch.

- Similarly, search for those lines containing **Section-A** and write the results into a file named **Section-A**.csv. Pipe it with a 3rd command to fetch and display the number of lines of this newly created file (use **wc** command and its relevant switch). Now, display the first 5 lines of **Section-A**.csv using **head** command and its relevant switch.

- Search within these 3 files (Section-A.csv, Section-B.csv, Section-C.csv) for those lines which **do not** contain **Muhammad** and create a file named **Muhammad-Excluded.csv**. Pipe it with a 3rd command to fetch and display the number of lines of this newly created file (use **wc** command and its relevant switch). Now, display the first 5 lines of **Muhammad-Excluded.csv** using **head** command and its relevant switch.

- That's it.

Submission Guidelines

- Take screenshots of your work and submit a pdf document of the screenshots. You can use any online tool for creating pdf from these screenshots.